

4(a) Since the tube is in equilibrium,

Weight of (tube and ball bearings) = Upthrust [1]

$$(m + M)g = AH\rho g$$

$$H = \frac{(M + m)}{\rho A} \quad [1]$$

(b) When the tube is displaced downwards by y , the upthrust will increase to $A(H+y)\rho g$ while the weight would remain the same as $(m+M)g$.

Hence the net force = $(m+M)g - A(H+y)\rho g$

$$= (m+M)g - AH\rho g - Ay\rho g$$

$$= -A\rho g y \quad \text{since } (m + M)g = AH\rho g$$

(negative means that the net force is opposite in direction to the displacement y) [1]

By N2L $(m + M)a = -A\rho g y$ [1]

$$a = -\frac{A\rho g}{m + M} y$$

- (c) From (b) since $a = -\left(\frac{\rho Ag}{M+m}\right)y$, thus $a \propto -y$. Hence motion is S.H.M. [1]

Comparing with defining equation: $a = -\omega^2 y$, then:

[1]

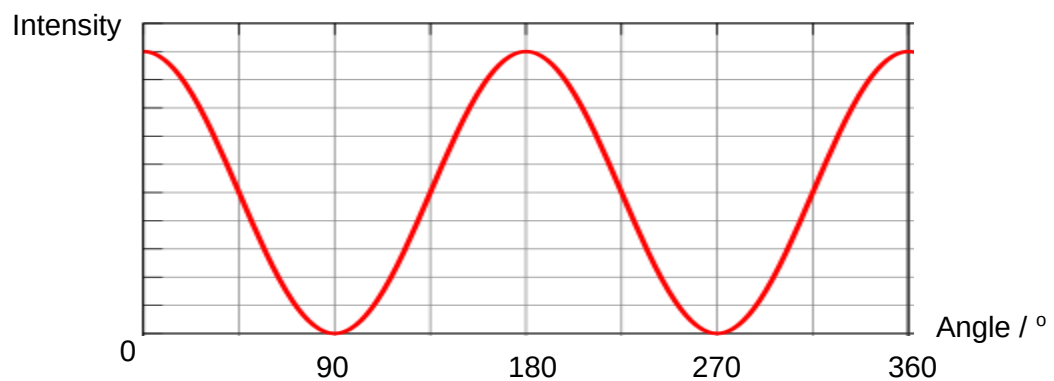
$$\begin{aligned}\omega^2 &= \frac{\rho Ag}{M+m} \\ \omega &= \sqrt{\frac{\rho Ag}{M+m}} \\ \frac{2\pi}{T} &= \sqrt{\frac{\rho Ag}{M+m}} \\ T &= 2\pi \sqrt{\frac{M+m}{\rho Ag}} \\ &= 2\pi \sqrt{\frac{0.012 + 0.025}{1.00 \times 10^3 \times 6.0 \times 10^{-4} \times 9.81}} \\ &= 0.4982 \\ &= 0.50 \text{ s (Shown)}\end{aligned}$$

[1]

- (d) As the test-tube oscillates, it experiences drag force exerted by the water.

This results in light damping and energy is gradually lost as heat. [1]

- (e)(i)



Sinusoidal graph [1]

Label period 3.3 s [1]

- (ii) The amplitude of oscillation is small because the frequency of the driving force (the waves) is too low (0.30 Hz) compared to the natural frequency of the test-tube (2 Hz). [1]

The amplitude of the oscillations can be increased by adding ball bearings to the test-tube to decrease the natural frequency so that it is closer to the frequency of the driving force. [1]

E(a) The direction of the induced emf is such as to produce an effect that opposes the